DS7400 Homework 3 Experiment Summary

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I investigated separate signal and context branches with a learned gate. The initial design underperformed, but adding context dropout and weight decay raised mean test accuracy from 93.83% for the controlled baseline to 94.77%.

1 Prediction task and baseline

The data contain cardiac point clouds and a U_heart electrical signal for each point. The task is to predict activation within the next 20 frames. Activation is defined as the frame with the largest positive signal change. The full trace generates labels; model inputs contain only current and past information. Each sample has 40 past signal values, XYZ, and normalized current time, giving 44 features standardized using training statistics.

44 inputs → 64 units + ReLU → 32 units + ReLU → 2 output logits

This HW2 baseline mixes both feature groups in its first layer. It uses cross-entropy loss and Adam.

2 Original hypothesis and proposed architecture

My original hypothesis was that signal history and spatial/time context have different meanings, so separate feature extraction followed by learned weighting might help. The dual-branch ablation maps 40 signal values to 32 ReLU features s and the four context values to 32 ReLU features c. Concatenation feeds a 64 → 32 ReLU → 2 classifier. The full design adds a 32-dimensional gate:

$$g = sigmoid(W_g[s, c] + b_g)$$

$$Classifier\ input = [2g \odot s, 2(1 - g) \odot c]$$

Products are elementwise. Zero gate weights and biases initialize g to 0.5; the factor of two matches the ungated feature scale. Dual-branch and gated models have 3,618 and 5,698 parameters. Shared branches and classifiers initialize identically within each seed. Possible failures include separating useful interactions too early, harder optimization, and learning case-specific position/time shortcuts. Parameter-count differences are another possible explanation.

3 Controlled experimental settings

Cases 0001–0003 train the models, case 0005 validates them, and case 0004 tests them. No case crosses splits. Identical sampling produces 116,748 training, 39,240 validation, and 42,964 test windows. Model seeds are 7400, 7401, and 7402; data sampling is fixed, and minibatch order is controlled separately from initialization. Batch size is 256. Each run selects the lowest-validation-cross-entropy checkpoint and uses a fixed 0.5 classification threshold.

The first suite used 20 epochs at learning rate 0.0001. Follow-ups compared 60 epochs at 0.0001 and 0.00003, then weight decay, 25%/50% context dropout, and a signal residual path. Follow-up settings were selected using validation results, without choosing parameters from new test scores.

4 Initial failures and subsequent modification

The first suite averaged 93.65% test accuracy for the baseline, 90.59% for dual branches, and 91.28% for gated fusion. This did not support the original hypothesis. Training accuracy was approximately 95%–96%, but some seeds produced many test false positives. Dual-branch seed 7400 reached 95.47% validation accuracy but only 88.78% test accuracy, with 4,810 false positives, 11 false negatives, and AUROC 0.9888. Strong ranking therefore did not ensure that the fixed decision boundary transferred across cases.

Some validation losses were still falling at epoch 20. However, gated seed 7401 showed falling training loss and worsening validation loss after epoch 3. More epochs helped only modestly, and a lower learning rate did not consistently resolve instability. Gate statistics did not show universal saturation, so “stuck gates” were not a sufficient explanation.

I then hypothesized excessive reliance on coordinates/time. I added weight decay and randomly removed the entire context embedding per training sample. The selected model uses weight decay 0.001, context dropout 0.5, learning rate 0.0001, and 60 epochs. Retained context embeddings are scaled by two; dropout is disabled at evaluation. This modification adds no model parameters.

5 Final results and analysis

After fixing the selected configuration, I evaluated all three seeds on case 0004. The comparison below uses the same 60-epoch budget and validation-based checkpoint selection for every architecture.

Model	Test accuracy	F1	AUROC	Accuracy SD
Baseline	93.83%	0.9374	0.9884	0.57 pp
Original gated	89.32%	0.9038	0.9883	2.85 pp
Gated with regularization	**94.77%**	**0.9500**	**0.9918**	**0.21 pp**

Values are means over three seeds; SD is sample standard deviation, and pp means percentage points. Final accuracies were 94.84%, 94.54%, and 94.95%, each above its paired baseline. Mean accuracy rose by 0.95 pp; error fell from 6.17% to 5.23%, with lower seed variability.

Weight decay reduced mean validation loss, and 50% context dropout improved it further; the signal residual variant did not outperform this configuration. This supports context regularization as useful, but does not prove a specific shortcut mechanism. The final model still trades lower precision for higher recall, so false positives remain.

6 Lessons limitations and next steps

A plausible fusion architecture did not automatically improve generalization. In these experiments, limiting context use helped more than extra epochs or a smaller learning rate. Conclusions are limited to three seeds and one case split: correlated points/windows do not replace independent cases. Early test results informed later investigation, so final testing is exploratory rather than a fresh untouched evaluation. Runtime differences across nodes are not a controlled efficiency comparison. New case splits and a coordinate/time-removal ablation would test generalization and the shortcut hypothesis.